A. P. VIMALADHITHYA

Junior Data Scientist | Machine Learning Engineer

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PROFESSIONAL SUMMARY

Data Science Consultant with hands-on experience in Machine Learning, Deep Learning, Statistical Analysis, and Predictive Modeling, converting structured data into actionable insights. Proficient in Python, SQL, TensorFlow, scikit-learn, Pandas, and NumPy, with strong skills in Exploratory Data Analysis (EDA), Feature Engineering, Data Preprocessing, Hypothesis Testing, and Data Visualization. Delivered a Credit Risk Prediction Model using **XGBoost** on 50K+ records (96.36% accuracy, AUC 0.85, Gini 35.97) and an Earthquake Damage Prediction classifier, and built AI-driven solutions including a multi-agent Agriculture Advisor using **CrewAI** and **Gemini API** and an **NLP**-based ATS Resume Analyzer. Seeking a Data Scientist role to develop machine learning algorithms and data-driven solutions that deliver business impact.

TECHNICAL SKILLS

Programming & Databases: Python, SQL (MySQL), Object-Oriented Programming
Machine Learning Algorithms: Predictive Modeling, Regression, Classification, Clustering, XGBoost, LightGBM, Random Forest, SVM, scikit-learn, Feature Engineering, Hyperparameter Tuning, Cross-Validation, Model Evaluation
Deep Learning & CV: TensorFlow, CNN, ANN, Transfer Learning
Statistics & Data Analysis: Statistical Analysis, Hypothesis Testing, Exploratory Data Analysis (EDA), Data Manipulation, Data Cleaning, Data Preprocessing, Data Processing, Pandas, NumPy
Data Visualization: Matplotlib, Seaborn, Power BI, Excel
AI & GenAI: Large Language Models (LLMs), Prompt Engineering, AI Agents
Backend: REST APIs, JSON, HTTP Methods (GET, POST, PUT, DELETE), API Integration
DevOps & Deployment: Docker(basic), Git, GitHub, Streamlit
Cloud & AI: AWS EC2/S3 (basic), Text Preprocessing, Tokenization (basic), Natural Language Processing (NLP), CrewAI, Gemini API

WORK EXPERIENCE

Data Science Consultant | Rubixe, Bangalore *Sep 2025 – Apr 2026*

- Delivered Credit Risk modeling using XGBoost on 50K+ structured financial records, achieving 96.36% accuracy, AUC 0.85, and Gini score of 35.97.
- Built an Earthquake Damage Prediction classification model using XGBoost (best model), achieving an accuracy of 71.3%, Precision of 71.5%, Recall of 71.3%, and F1-score 0.702 on seismic data.
- Performed data cleaning, preprocessing, feature engineering, and EDA on large-scale structured datasets using Python, Pandas, and NumPy.
- Optimized model performance through hyperparameter tuning and cross-validation; evaluated using Accuracy, AUC, Gini, Precision, Recall, and F1-score metrics.

Technologies: *Python, Pandas, NumPy, scikit-learn, XGBoost, LightGBM, Matplotlib, Seaborn*

KEY PROJECTS

AI-Powered Agriculture Advisor — CrewAI Multi Agent Web Application *Aug 2026*

GitHub: github.com/vimaladhithya/AI-Powered-Agriculture-Advisor
Live Demo: AI-Powered Agriculture Advisor App

- Developed and deployed a multi-agent AI agricultural advisory system using CrewAI and Gemini LLM for data-driven farming recommendations.
- Designed Location, Soil, and Weather Agents to analyze geospatial, soil, and climate data for crop suitability assessment.
- Integrated agricultural datasets and real-time environmental data to generate personalized crop recommendations.
- Automated farming advisories for planting, irrigation, harvesting, and weather-risk management to support informed decision-making.

Technologies: *Python, CrewAI, Gemini API, Streamlit, Pandas, REST APIs*

ATS Score and Resume Optimizer — NLP & AI-Powered Web Application *June 2026*

GitHub: github.com/vimaladhithya/ATS-Score-and-Resume-Optimizer
Live Demo: ATS Resume Analyzer App

- Developed and deployed an ATS Resume Analyzer to evaluate resume compatibility with job descriptions and technical skill requirements.
- Implemented skill extraction, keyword matching, and ATS scoring to identify skill gaps and improve resume relevance.
- Integrated Sentence Transformers for semantic similarity analysis between resumes and job descriptions.
- Built an AI-powered resume optimization module using Gemini API to generate ATS-friendly, role-specific resume recommendations.

Technologies: *Python, Streamlit, NLTK, Sentence Transformers, Google Gemini API, ReportLab*

Credit Risk Prediction — (Client Project) *Feb – Mar 2026*

GitHub: github.com/vimaladhithya/credit-risk-prediction

- Designed a Credit risk prediction model to classify loan defaulters using financial, demographic, and transactional data from 50K+ records.
- Conducted data cleaning, preprocessing, feature engineering, and EDA across three structured tables — Enquiry, Customer Account, and Demographic — to improve data quality and model readiness.
- Implemented and compared Random Forest, LightGBM, and XGBoost; applied hyperparameter tuning to optimize each model's performance.
- Achieved the best performance with XGBoost: Accuracy 96.36%, Gini 35.97, AUC 0.85.

Technologies: *Python, Pandas, NumPy, scikit-learn, XGBoost, LightGBM, Matplotlib*

PROFESSIONAL TRAINING

Certified Data Scientist Program — Datamites *Jul – Sep 2025*

- Completed intensive training in Machine Learning, Deep Learning, Statistics, Python, and SQL with hands-on end-to-end project work.
- Built and evaluated ML models on real-world datasets with practical experience in data preprocessing, feature engineering, and model evaluation.

CERTIFICATIONS

- Certified Data Scientist (CDS) — IABAC, 2026
- Data Science Foundation (DSF) — IABAC, 2026
- Certified Data Scientist — NASSCOM (Gold, 91%), 2026
- Certified Data Scientist Program — Machine Learning, Statistics & Deep Learning — Datamites, 2025
- SQL with PostgreSQL & pgAdmin — Udemy, 2025
- AI Tools Workshop — Be10x, 2025

EDUCATION

Bachelor of Engineering (Computer Science) *2021 – 2025*

Mepco Schlenk Engineering College, Sivakasi